Status	Finished
Started	Monday, 2 November 2026, 10:46 PM
Completed	Monday, 2 November 2026, 10:54 PM
Duration	7 mins 57 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

Which of the following is not a limitation of binary search algorithm?

a. Requirement of sorted array is expensive when a lot of insertion and deletions are needed

b. Must use a sorted array

c. Binary search algorithm is not efficient when the data elements more than 1500 ✓

d. There must be a mechanism to access middle element directly

Question **2**

Correct

Mark 1.00 out of 1.00

_____ search takes a sorted/ordered list and divides it in the middle.

a. Linear

b. Hash

c. Both (1) & (3)

d. Binary ✓

Question **3**

Correct

Mark 1.00 out of 1.00

What is the time complexity of linear search in the worst case?

a. O(1)

b. O👎 ✓

c. O(log n)

d. O(n log n)

Question **4**

Correct

Mark 1.00 out of 1.00

In linear search, how is the element searched?

a. By sorting the list first
b. By comparing each element in the list sequentially ✓
c. By dividing the list into halves
d. By using a hash function

Question **5**

Correct

Mark 1.00 out of 1.00

What is the best-case time complexity of linear search?

a. O(log n)
b. O(n log n)
c. O👎
d. O(1) ✓

Question **6**

Correct

Mark 1.00 out of 1.00

Which of the following is not the required condition for a binary search algorithm?

a. The list must be sorted
b. There should be direct access to the middle element in any sublist
c. Number values should only be present

Number values should only be present

d. There must be a mechanism to delete and/or insert elements in the list ✓

Question **7**

Correct

Mark 1.00 out of 1.00

Finding the location of a given item in a collection of items is called

a. Mining

b. Discovering

c. Searching ✓

d. Finding

Question **8**

Correct

Mark 1.00 out of 1.00

What type of search would be most appropriate for finding an element in a list that is frequently updated?

a. Binary search

b. Interpolation search

c. Linear search ✓

d. Hash search

Question **9**

Correct

Mark 1.00 out of 1.00

_____sort is the simplest sorting algorithm that works by repeatedly swapping the adjacent elements in case they are unordered in n-1 passes.

a. Bubble ✓

b. Insertion

c. Complexity

d. Selection

Question **10**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

a. n/2
b. n ✓
c. n-1
d. log n

Question **11**

Correct

Mark 1.00 out of 1.00

In binary search, how is the middle element determined?

a. By using a hash function
b. By starting from the first element
c. By dividing the list length by two ✓
d. By comparing each element sequentially

Question **12**

Correct

Mark 1.00 out of 1.00

In which situation is linear search more efficient than binary search?

a. When the list is large and sorted
b. When the list is small and unsorted ✓
c. When the list is large and unsorted
d. When the list is small and sorted

Question **13**

Correct

Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

a. 1000
b. 100
c. 500
d. 10 ✓

Question **14**

Correct

Mark 1.00 out of 1.00

In _____ checks the elements of a list, one at a time, without skipping any element.

- a. Linear search ✓
- b. Binary search
- c. Hash search
- d. Both (1) & (3)

Question **15**

Correct

Mark 1.00 out of 1.00

What happens when the element is found in linear search?

- a. The search backtracks to find duplicate elements
- b. The search stops immediately ✓
- c. The search continues until the end of the list
- d. The search starts over from the beginning